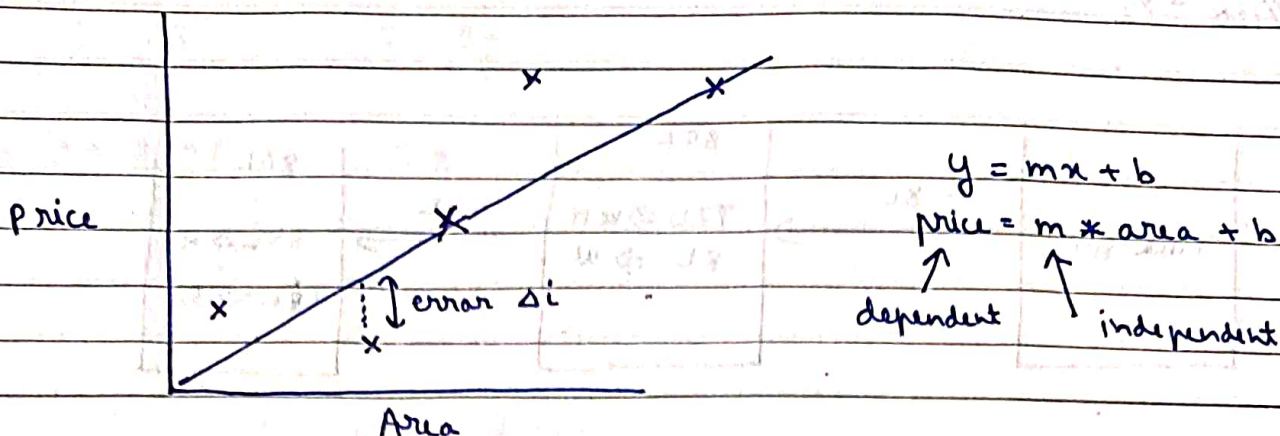


① Linear Regression



We draw multiple lines and calculate the sum of error of the points for each line. The line for which the sum of errors is least that line is chosen as the best fit line.

$$\frac{1}{n} \sum_{i=1}^n (\Delta_i)^2 \Rightarrow \text{Sum of errors for each points for a line is calculated using this formula.}$$

```
plt.xlabel('area (sq ft)')
plt.ylabel('price (US $)')
plt.scatter(df.area, df.price, color='red', marker='+')
```

```
reg = linear_model.LinearRegression()
```

```
reg.fit(df[['area']], df.price)
```

```
reg.predict(3300)
```

```
reg.coef_
```

```
reg.intercept_
```

It is calculated internally to choose the best fit line. With this slope and intercept value we calculate the predicted value using the formula

$$y = mx + c$$

```
plt.plot(df.area, reg.predict(df[['area']]), color='blue')
```